1) A finite wing of span b has a circulation distribution:

$$\Gamma(y) = \Gamma_0 \left[1 - \left(\frac{2y}{b}\right)^2\right] \qquad -b/2 \le y \le b/2$$

w/ freestream velocity V_∞ & density ρ. Use Prandtl's lifting line theory.

a) Find the lift

b) Find the trailing shed vortex sheet strength

c) Find the downwash along the lifting line

d) Find induced drag. Set up the integral. Don't need to evaluate.

a) $L = \rho V_\infty \int_{-b/2}^{b/2} \Gamma(y)\, dy = 2\Gamma_0 \int_0^{b/2} \left[1 - \frac{4y^2}{b^2}\right] dy$

$$2\Gamma_0 = \int_0^{b/2} \left[1 - \frac{4y^2}{b^2}\right] dy = 2\Gamma_0 \left[y - \frac{4y^3}{3b^2}\right]_0^{b/2}$$

$$= 2\Gamma_0 \left[\frac{b}{2} - \frac{4\left(\frac{b}{2}\right)^3}{3b^2}\right] = \Gamma_0 \frac{2}{3}\left[\frac{b}{2} - \frac{b}{6}\right] = \frac{2}{3}\Gamma_0$$

$$\boxed{L = \frac{2}{3}\rho V_\infty \Gamma_0 b}$$

b)



$$\gamma = -\frac{dP}{dy} \rightarrow \gamma(y) = -\frac{dP}{dy}\,\Gamma_0\left[1 - \frac{4y^2}{b^2}\right]$$

$$= \boxed{\frac{8\Gamma_0}{b^2}\,y}$$

c) $$w(y_0) = -\frac{1}{4\pi}\int_{-b/2}^{b/2} \frac{\gamma(y)}{y_0 - y}\,dy$$

$$= \int_{-b/2}^{b/2} \frac{y}{y^2 - x^2}\, dy = -\ln|y_0 - y| \Big|_{-b/2}^{b/2} = -\ln|y_0 - \tfrac{b}{2}| + \ln|y_0 + \tfrac{b}{2}|$$

$$= \ln\left|\frac{y_0 + b/2}{y_0 - b/2}\right|$$

$$= \int_{-b/2}^{b/2} \frac{dy}{1 + \frac{x^2}{(y_0 - y)^2}} = -\int_{y_0 + b/2}^{y_0 - b/2} \frac{du}{1 + u^2} = -\tan^{-1} u \Big|_{y_0 + b/2}^{y_0 - b/2}$$

$$u = \frac{x}{y_0 - y} \qquad dy = -du$$

$$= \tan^{-1}\left(\frac{y_0 + b/2}{x}\right) - \tan^{-1}\left(\frac{y_0 - b/2}{x}\right)$$

$$= \tan^{-1}\left(\frac{\frac{y_0 + b/2}{x} - \frac{y_0 - b/2}{x}}{1 + \frac{y_0 + b/2}{x}\cdot\frac{y_0 - b/2}{x}}\right) = \tan^{-1}\left(\frac{bx}{x^2 + y_0^2 - b^2/4}\right)$$

$$\frac{dB_x}{dy} = \frac{\mu_0 I}{2\pi}\, \frac{d}{dy}\left[\tan^{-1}\left(\frac{bx}{x^2 + y_0^2 - b^2/4}\right)\right]$$

$$= \frac{\mu_0 I}{2\pi}\, \frac{1}{1 + \left(\frac{bx}{x^2 + y_0^2 - b^2/4}\right)^2}\, \frac{d}{dy}\left(\frac{bx}{x^2 + y_0^2 - b^2/4}\right)$$

$$= \frac{\mu_0 I}{2\pi}\, \frac{(x^2 + y_0^2 - b^2/4)^2}{(x^2 + y_0^2 - b^2/4)^2 + b^2 x^2}\, \frac{-bx(-2y)}{(x^2 + y_0^2 - b^2/4)^2}$$

$$= \frac{\mu_0 I}{2\pi}\, \frac{2bxy}{x^4 + 2x^2(y_0^2 - b^2/4) + (y_0^2 - b^2/4)^2 + b^2 x^2}$$

$$= \frac{\mu_0 I}{2\pi}\, \frac{2bxy}{x^4 + 2x^2(y_0^2 + b^2/4) + (y_0^2 - b^2/4)^2}$$

$$B_x = \frac{\mu_0 I}{2\pi}\, 2bxy \int_{-b/2}^{b/2} \frac{dz}{x^4 + 2x^2(y_0^2 + b^2/4) + (y_0^2 - b^2/4)^2}$$

$$= \frac{\mu_0 I}{2\pi}\, 2bxy \int_{-b/2}^{b/2} \frac{dz}{x^4 + 2x^2(y_0^2 + b^2/4) + (y_0^2 - b^2/4)^2}$$

d) $D_c = -\rho V_\infty \Gamma \alpha_i \qquad \alpha_i = -\frac{w}{V_\infty}$

$$D_c = -\rho \int_{-b/2}^{b/2} \Gamma(y)\, w(y)$$

$$D_c = -\rho \int_{-b/2}^{b/2} \Gamma_0 \left[1 - \left(\frac{2y}{b}\right)^2\right] \left[-b + y_0 \ln\left|\frac{y_0 + b/2}{y_0 - b/2}\right|\right] dy$$

2) A thin, symmetric airfoil w/ chord $c = 0.5\ m$ starts from rest & accelerates to $V_\infty = 20\ m/s$ in air. It is held at $\alpha = 5^\circ$

a) Find the circulation Γ around the airfoil.

b) Find the strength of the starting vortex and explain why it must exist

c) Find L' & C_ℓ

d) Find M'_{LE}, $C_{m,LE}$ & x_{cp}

e) Find vortex sheet strength γ at mid chord & at quarter-chord, what happens to γ as $x \to 0$ and is it physical

a) $\Gamma = \pi \alpha c V_\infty = \pi (0.0873)(0.5)(20)$

$= 2.74 \frac{m^2}{s}$

b) $-2.74\ m^2/s$

The fluid started at rest, so the curve enclosing both the airfoil and the wake had $\Gamma = 0$. By Kelvin's circulation theorem it must still have $\Gamma = 0$. The airfoil then sheds a starting vortex w/ exact & opposite strength

c) $L' = \rho V_\infty \Gamma = 1.2 \cdot 20 \cdot 2.74 = 65.8 \frac{N}{m}$

$C_\ell = 2\pi\alpha = 0.548$

d) $M'_{LE} = -\rho V_\infty^2 c^2 \frac{\pi}{4}\alpha = -(1.2)(400)(0.25)\frac{\pi(0.0875)}{4}$

$= -8.22 \frac{N \cdot m}{m}$

$C_{M_{LE}} = -\frac{C_\ell}{4} = -0.137$

e) $\gamma(\theta) = 2\alpha V_\infty \frac{1+\cos\theta}{\sin\theta}$

$x = \frac{c}{2}(1-\cos\theta)$

For $x = \frac{c}{2}$

$\frac{c}{2} = \frac{c}{2}(1-\cos\theta)$

$1 = 1-\cos\theta \rightarrow 0 = -\cos\theta \rightarrow \theta = 90^\circ$

$2\alpha V_\infty \frac{1+\cos(90)}{\sin(90)} = 3.49 \text{ m/s}$

$x = \frac{c}{4} \rightarrow \theta = 60^\circ = \gamma = 6.05 \text{ m/s}$

When x is approaching 0, θ becomes 0 t

in $r(\theta) = 2\alpha V_0 \frac{1+\cos\theta}{\sin\theta}$, $\sin\theta$ will approach

0 so r will approach ∞ which is

not physical

3) A cambered airfoil is tested in a wind tunnel, with these results

α	C_l	$C_{m_{c/4}}$
2°	0.44	-0.045
8°	1.08	-0.0039

a) Find a_0 and m_0

b) Find x_{ac} and $C_{m_{ac}}$. Check that $C_{m_{ac}}$ is the same at both angles.

$$a_0 = \frac{dC_l}{d\alpha} = \frac{1.08 - 0.44}{6^\circ} = 0.1067/\text{deg} = 6.11/\text{rad}$$

$$m_0 = \frac{dC_m}{d\alpha} = \frac{-0.0039 - (-0.045)}{6^\circ} = 0.001/\text{deg}$$

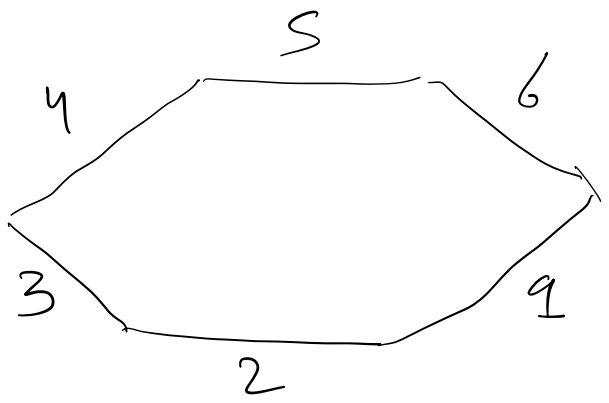
b) $x_{ac} = -\frac{m_0}{a_0} + 0.25 = -\frac{0.001}{0.1067} + 0.25 = 0.241$

$$C_{m_{ac}} = C_{m_{c/4}} + C_l (x_{ac} - 0.25)$$

$2^\circ \rightarrow -0.045 + 0.44(-0.0094) = -0.0491$ ✓

$8^0 \rightarrow -0.039 + 1.08(-0.0094) = -0.0491$ ✓

4) A coarse 6-panel solution for 12% thick airfoil gives the values



n	γ	S
1	-17	0.16
2	-18.8	0.16
3	-6	0.17
4	20.5	0.17
5	21	0.18
6	17	0.16

a) Check the Kutta condition

b) Find Γ, L' if $V_\infty = 20$

c) $\gamma_6 + \gamma_1 = -17 + 17 = 0$

b) $\Gamma = \sum \gamma_j s_j = -17(0.16) + -18.8(0.16) + -6(0.17) + 20.5(0.17) + 21(0.18) + 17(0.16)$

$= 2.86 \ m^2/s$

$L' = 70.07 \ N/m$